

Experiment : 9

Orchestrating Serverless Functions with AWS Step Functions

What is AWS Step Functions?

It is a service by Amazon Web Services that helps you:

- Connect multiple serverless functions
- Control execution order
- Handle errors and retries
- Visualize workflows

Basic Components

✓ State Machine

- Main workflow
- Written in JSON

✓ States (Steps in workflow)

- **Task** → calls AWS Lambda
- **Choice** → decision (if/else)
- **Wait** → delay
- **Parallel** → run tasks together
- **Pass** → pass data
- **Fail / Succeed** → end

2. Architecture Flow

Flow:

Start → Task → Decision → Next Step → End

3. Step-by-Step Working

Define Workflow

- Create state machine in JSON

Trigger Execution

- API / Event / Manual

Execute First State

- Calls Lambda function

Data Passing

- Output → next input

Decision Making

- Choice state (if/else)

Parallel Execution (Optional)

- Run tasks together

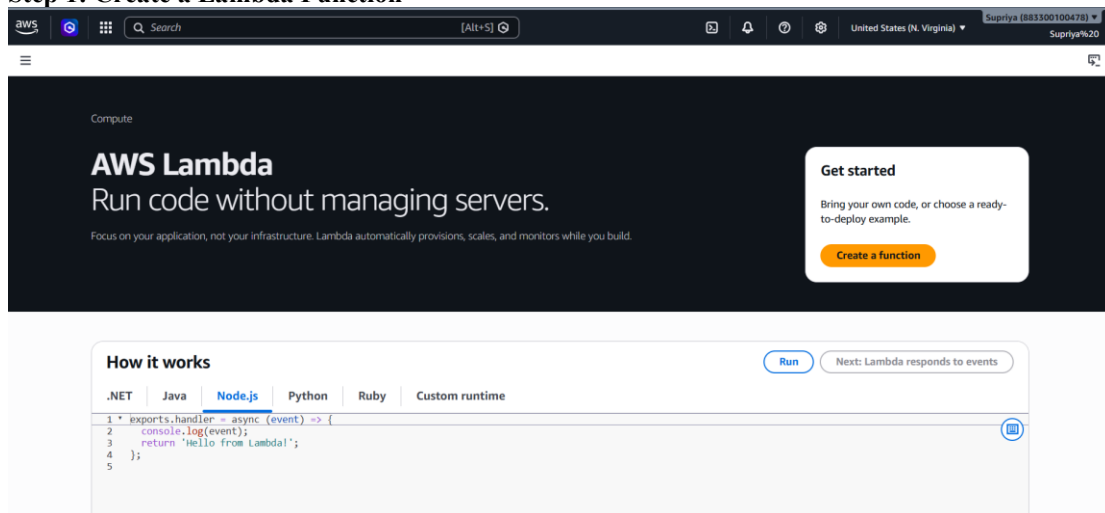
Error Handling

- Retry, timeout, catch errors

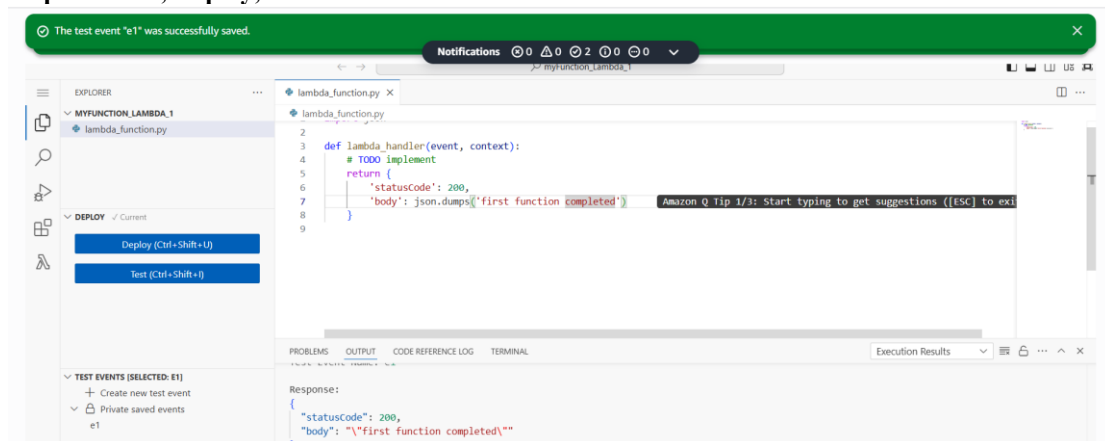
End Execution

- Success  or Fail 

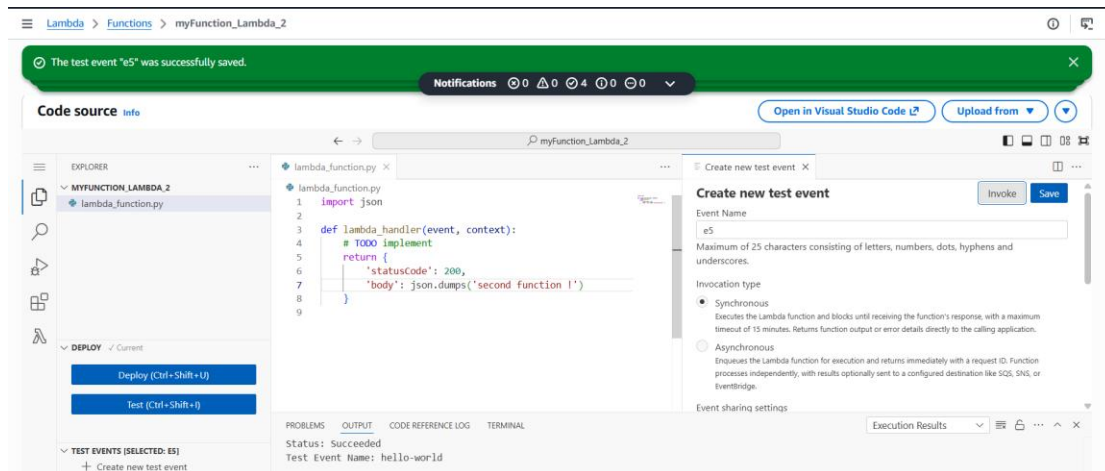
Step 1: Create a Lambda Function



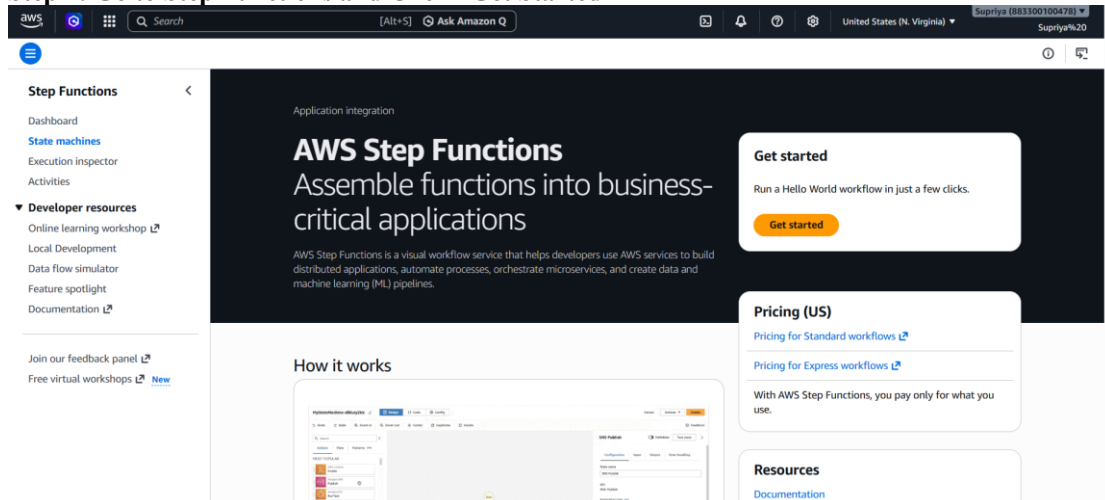
Step 2: Write, Deploy, and Test the Function



Step 3: Create One More Function, Write, Deploy and Test



Step 4: Go to Step Functions and Click “Get Started”



Step 5: Create State Machine and Add Functions

- Click on State Machine in AWS Step Functions
- Click Create State Machine
- Choose Workflow Studio (visual editor)
- Drag and drop Invoke Function (Task)
- Select your AWS Lambda function
- Configure input/output if needed
- Click Create and then Confirm

Step Functions

Dashboard

State machines

Execution inspector

Activities

Developer resources

Online learning workshop

Local Development

Data flow simulator

Feature spotlight

Documentation

Join our feedback panel

Free virtual workshops

State machine successfully created

Notifications

Machine1

Design

Code

Config

Exit

Actions

Execute

Save

Undo

Redo

Zoom in

Zoom out

Center

Duplicate

Delete

Feedback

Workflow

Definition

The top level state machine properties for this workflow. [Learn more](#)

State machine query language

JSONata

Start at

The state that is the starting point of the workflow.

Lambda Invoke

Comment - optional

A human-readable description of the state machine.

A description of my state machine

```
graph TD; Start((Start)) --> LambdaInvoke[Lambda: Invoke  
Lambda Invoke]; LambdaInvoke --> LambdaInvoke1[Lambda: Invoke  
Lambda Invoke (1)]; LambdaInvoke1 --> End((End))
```

Step 6: Click on Execute

Step Functions

Dashboard

State machines

Execution inspector

Activities

Developer resources

Online learning workshop

Local Development

Data flow simulator

Feature spotlight

Documentation

Join our feedback panel

Free virtual workshops

Execution started successfully

Execution: cc04a556-03ab-48d8-a1c6-44bd7492583b

Edit state machine

New execution

Actions

Details

Execution input and output

Definition

Execution status

Succeeded

Execution type

Standard

Execution ARN

arn:aws:statesus-east-1:883300100478:execution:Machine1:cc04a556-03ab-48d8-a1c6-44bd7492583b

IAM role ARN

arn:aws:iam::883300100478:role/service-role/StepFunctions-Machine1-role-dugr1q2qq

State transitions

4

Start time

Apr 24, 2026, 07:36:02.146 (UTC+05:30)

End time

Apr 24, 2026, 07:36:02.993 (UTC+05:30)

Duration

00:00:00.847

Alias

-

Version

-

Graph view

Table view

Step Functions

Dashboard

State machines

Execution inspector

Activities

Developer resources

Online learning workshop

Local Development

Data flow simulator

Feature spotlight

Documentation

Join our feedback panel

Free virtual workshops

Graph view

Actions

Step details

Choose a step to view its details.

```
graph TD; Start((Start)) --> LambdaInvoke[Lambda: Invoke  
Lambda Invoke]; LambdaInvoke --> LambdaInvoke1[Lambda: Invoke  
Lambda Invoke (1)]; LambdaInvoke1 --> End((End))
```

In progress

Failed

Caught error

Cancelled

Succeeded

Event view

State view